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Genotyping and molecular analysis of *Enterocytozoon bieneusi* isolated from immunocompromised patients in Iran

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ABSTRACT

Microsporidia are known as opportunistic unicellular pathogens, particularly so in individuals with congenital or acquired immunodeficiency. *Enterocytozoon bieneusi* is one of the most common species infecting both immunocompromised and immunocompetent individuals. The aim of this study was to assess the distribution of *E. bieneusi* genotypes among immunocompromised patients in Iran. From 329 stool samples referred for parasitological analysis during 2011–2014, 14 samples from immunocompromised patients proving positive for *E. bieneusi* by SSU rDNA analysis were selected. Genotyping was carried out using specific primers targeting the Internal Transcribed Spacer (ITS) region. Subsequently, all samples were sequenced and results queried against the GenBank database. Moreover, sequences were subject to phylogenetic analysis. The expected amplification product was generated for all samples. Genotype D was identified in patients with HIV+/AIDS, transplant recipients, and cancer patients, while Genotype E was identified only in cancer and HIV+/AIDS patients. Phylogenetic analysis revealed that there was no relationship between genotypes and types of immunosuppression, whereas most genotype D isolates grouped with those described previously from cattle, horses, birds, and humans. *E. bieneusi* genotype D appears to be the most frequent genotype in immunocompromised patients, while Genotype E was observed only in HIV+/AIDS patients and cancer patients, not transplant recipients.

1. Introduction

Microsporidia are unicellular, obligate intracellular pathogens that infect a broad spectrum of vertebrates and invertebrates almost all over the world. Microsporidia are known as an opportunistic pathogen, particularly in individuals with deficient immune systems, including acquired immune deficiency syndrome (AIDS) (Batman et al., 2007; Lono et al., 2011; Shadduck, 1989), transplant recipient cancer patients undergoing immunosuppressive treatment, and/or other patients with acquired or congenital immunity system failure (Anane and Attouchi, 2010; Didier and Weiss, 2006; Lono et al., 2008; Teachey et al., 2004). Phylogenetic analysis of critical genes supports a close relationship between microsporidia and fungi (Didier and Weiss, 2006; Gill and Fast, 2006), and although microsporidia were recently reclassified as fungi, the relationship of this microorganism with fungi remains vague. Among human-infecting microsporidian species, *Enterocytozoon bieneusi* is one of the most prevalent species in immunocompromised patients with gastrointestinal disorders (Mathis et al., 2005; Weber et al., 1994). *E. bieneusi* was first reported in a Haitian AIDS patient (Desportes et al., 1985). Several studies have been carried out on the genetic diversity and the route of transmission of *E. bieneusi* due to its frequent comprehensive occurrence in immunocompromised and even immunocompetent individuals (Abreu-Acosta et al., 2005).

Based on intra-species DNA sequence variation in the ITS region, at least 93 genotypes of *E. bieneusi* have been described in a vast range of hosts, representing humans and vertebrate animals, with some genotypes having multiple and synonymous names. Some genotypes are host-specific, while others may infect a variety of host species. Until now, about 45 genotypes have been identified in humans, including 34 genotypes recognized only in humans, and 11 genotypes that have been reported in animals as well as in humans (Matos et al., 2012; Santin and Fayer, 2011). Several studies have established the fact that genotype distribution as well as predominance of genotypes varies according to geographical region and source of infection (Dengjel et al., 2001; Leelayoova et al., 2005). Undoubtedly, knowledge on the distribution of *E. bieneusi* genotypes is critical to identifying groups of individuals potentially being particularly susceptible to infection with certain

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genotypes. Moreover, genetic characterization of *E. bieneusi* leads to a better understanding of the route(s) of transmission and the sources involved in the transmission cycle.

However, in Iran, studies on microsporidiosis in susceptible groups have been initiated only recently, including prevalence studies in HIV+/AIDS patients, liver transplant recipient patients, and animals (Agholi et al., 2013a,b; Jamshidi et al., 2012; Mirjalali et al., 2014), with one study reporting on the existence and distribution of *E. bieneusi* genotypes in urban pigeons (Pirestani et al., 2013).

Here, we hypothesized that some *E. bieneusi* genotypes might be specific to certain types of immunity failures. Therefore, the aim of our study was to assess the distribution and phylogenic relationships of *E. bieneusi* genotypes among a variety of susceptible patient groups.

2. Materials and methods

2.1. Sampling

A total of 329 stool samples from patients with or without diarrhea, including 80, 234, and 15 HIV+/AIDS, cancer, and transplant recipient patients, respectively, were surveyed for microsporidia infection using both parasitology and molecular methods. HIV infection was confirmed by Western blotting, and AIDS was defined by a CD4+ T cell count less than 200 per μL of peripheral blood and receiving anti-retroviral treatment.

Microsporidia spores with sizes commensurate to those of *E. bieneusi* (1.0–1.5 μm) were identified in stool samples by trichrome-blue staining (Ryan et al., 1993), and for confirmation, the presence of *E. bieneusi* spores was proved in all isolates by amplification of a specific small subunit (SSU) rDNA fragment using already published primers (Mirjalali et al., 2014).

Finally, fourteen *E. bieneusi*-positive samples confirmed at SSU rDNA level from three susceptible groups including transplant recipients ($N = 4$), cancer patients ($N = 5$), and HIV+/AIDS patients ($N = 5$) with chronic diarrhea, were included in our study.

2.2. DNA extraction

DNA extraction was performed according to a protocol published elsewhere (Mirjalali et al., 2014). Briefly, 250 μL of stool suspended in PBS (pH = 7.5) was transferred to a 1.5-mL tube. After centrifuging at 10,000 rpm for 5 min and discarding the supernatant, 400 μL of lysis buffer (Tric 100 mM, EDTA 10 mM, SDS 2%, and 20 μg/μL of Proteinase K) and 300 μL/v acid-washed glass beads (425–600 μm) were added to the pellet. Tubes were mixed vigorously for 2 min and heated at 60 °C for 4 h. Samples were vigorously agitated every 30 min. Subsequently, samples were centrifuged at 3000 rpm for 5 min, and the supernatant was collected and processed using a commercially available Bioneer stool DNA extraction kit (Bioneer Corporation, Daejeon, Korea) according to the manufacturer's instructions. Purified DNA was stored at −20 °C until use.

Nested PCR were performed using a set of specific primers targeting an ITS fragment of *E. bieneusi*. Primers were designed using the online software Primer-BLAST (http://www.ncbi.nlm.nih.gov/tools/primer-blast/). First, PCR was carried out using EbGeno-fe (5′-TTC AGA TGG TCA TAG GGA TG-3′) as the forward primer and EbGeno-re (5′-ATT AGA GCA TTC CGT GAG G-3′) as the reverse primer, which together amplify a 465-bp specific fragment. Amplification was accomplished in final volume of 50 μL, containing 25 μL of 2X PCR master mix (Ampliqon, Denmark), 10 pM of each primer, and 5 μL of DNA using PeqLab thermocycler (PEQLAB Biotechnologie GmbH, Germany). The conditions for the PCR reaction were 95 °C for 5 min followed by 35 cycles of 94 °C for 40 s, 53 °C for 45 s, and 72 °C for 45 s, and final extension of 72 °C for 4 min. A second PCR was performed using the forward primer EbGeno-fi (5′-TCG GCT CTG AAT ATC TAT GG-3′) and the reverse primer EbGeno-ri (5′-ATT CTT TCG CGC TCG TC-3′) in order to amplify a 410-bp internal fragment used for genotype specification. The second PCR reaction used 95 °C for 5 min, followed by 30 cycles of 94 °C for 35 s, 55 °C for 40 s, and 72 °C for 40 s, plus a final extension step at 72 °C for 5 min.

Subsequently, 10 μL of each PCR product was electrophoresed on a 1.5% agarose gel in TBE (0.09 M Tris, 0.09 M boric acid, 2 mM EDTA) stained with 0.5 μg/mL ethidium bromide and visualized using a UV Transilluminator.

2.3. Phylogenetic analysis

PCR products were sequenced according to the Sanger method using an ABI 3730 sequencer (Bioneer, Daejeon, South Korea), and results were compared with sequences already deposited in the GenBank database, using BLAST software. For phylogenetic analysis, all sequences were aligned using the clustalW algorithm implemented in the Bio Edit software (Hall, 1999). The best DNA tree model was determined, and phylogenetic trees were developed using the Maximum-Likelihood algorithm and the Tamura 3-parameter model as implemented in the MEGA6 software (Tamura et al., 2013). Statistical support for distance was evaluated using bootstrapping (1000 replicates). Intra-genotype distance and similarity of the representative sequences were also calculated using MEGA6 software.

3. Results

Amplicons with the expected size of about 410 bp corresponding to the ITS fragment were obtained for all isolates. After sequencing and comparison of the data with those in GenBank, genotypes E and D were identified in 5 and 9 isolates, respectively. Genotype D was identified in patients with AIDS ($N = 3$), cancer patients ($N = 2$), and transplant recipients ($N = 4$), while Genotype E was identified in cancer ($N = 2$) and AIDS ($N = 3$) patients, but not in transplant recipients. All sequences generated in the present study showed a similarity higher than 99% to sequences already submitted to GenBank. Accession numbers and additional information on the samples are summarized in Table 1.

The genetic diversity across our *E. bieneusi* isolates, which was calculated by multiple alignment, showed specific signatures in the discriminative portion of the sequenced ITS fragment of each genotype (Fig. 1).

As evidenced by phylogenetic analysis, sequences belonging to each genotype segregated into specific clusters. The analysis showed that no relationship existed between genotype and types of immunity failure. Most Genotype D isolates grouped with those described previously from cattle, horses, birds, and humans. All Genotype E isolates of our study were placed in separate clusters (Fig. 2). Similarity among genotypes D and E in our study was 99.8% and 99.3%, respectively.

Table 1
Overview of *Enterocytozoon bieneusi* isolates included in the study.

Isolate	Type of immune deficiency	Sex	Accession no	Genotype
E. b. 4	HIV+/AIDS	Female	KJ700428	E
E. b. 11	HIV+/AIDS	Male	KJ700429	D
E. b. 14	Liver transplantation	female	KJ700434	D
E. b. 15	HIV+/AIDS	Male	KJ700430	D
E. b. 17	HIV+/AIDS	Female	KJ700431	D
E. b. 74	Breast cancer	Female	KJ700424	E
E. b. 92	Heart transplantation	Female	KJ700435	D
E. b. 117	Acute myeloid lymphoma (AML)	Male	KJ700425	D
E. b. 118	Gastric cancer	Male	KJ700426	E
E. b. 160	Kidney transplantation	Male	KJ700436	D
E. b. 165	HIV+/AIDS	Female	KJ700432	E
E. b. 173	HIV+/AIDS	Male	KJ700433	E
E. b. 186	Liver transplantation	Male	KJ700437	D
E. b. 324	Gastric cancer	Male	KJ700427	D

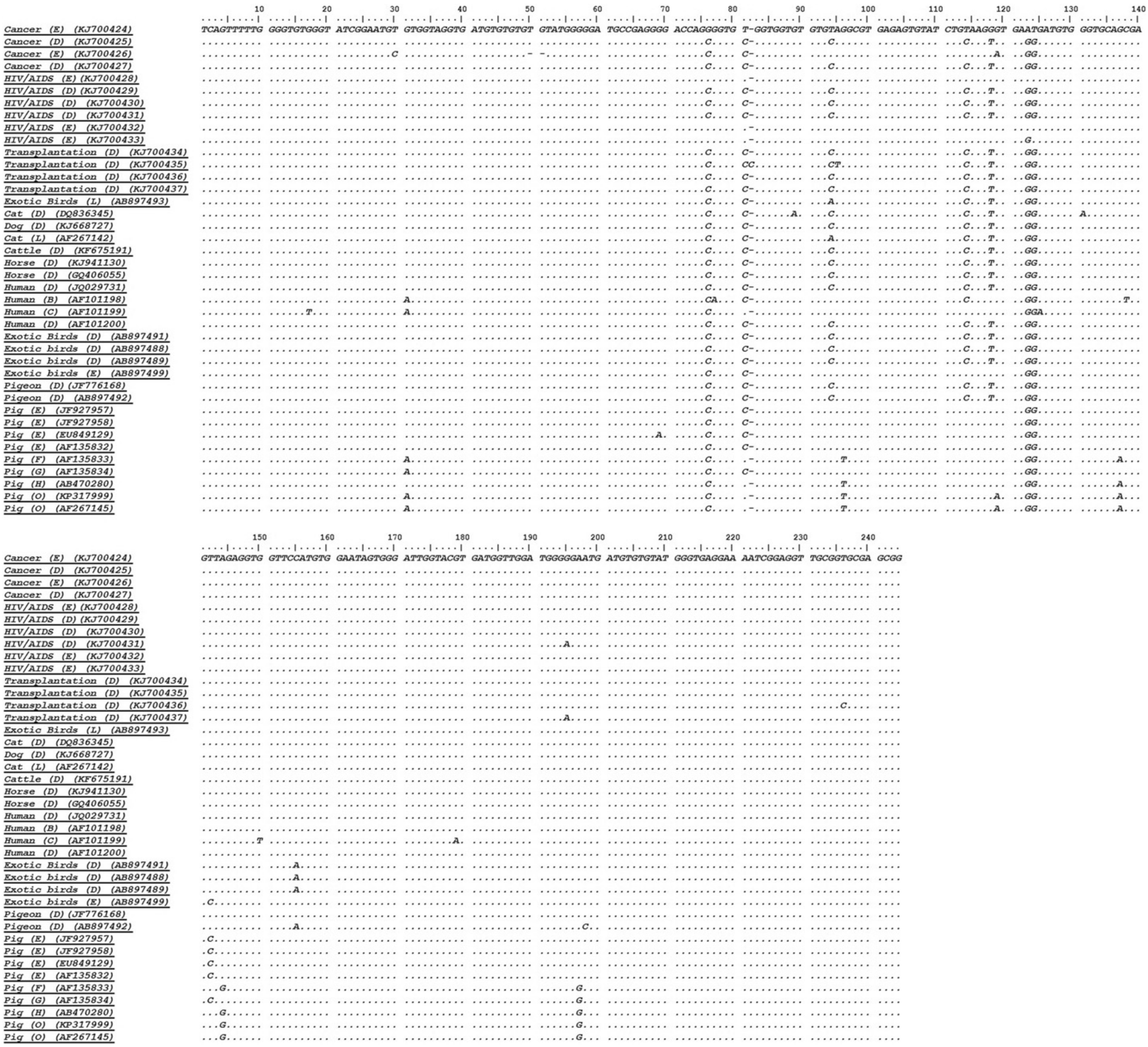

Fig. 1. Alignment of ITS nucleotide positions of study sequences and reference sequences retrieved from GenBank.

4. Discussion

E. bieneusi is the most prevalent microsporidial species isolated from immunocompromised and even immunocompetent persons (Abdelmalek et al., 2011; Abreu-Acosta et al., 2005; Chabchoub et al., 2012; Lejeune et al., 2005; Ojuromi et al., 2012). However, knowledge on infection sources, molecular epidemiology data, including data on routes of transmission are critical to controlling and preventing cases of microsporidiosis.

To date, several studies involving *E. bieneusi* in animals with close contact to humans have highlighted the potential for zoonotic transmission (Lee, 2000; Leelayoova et al., 2006; Pirestani et al., 2013; Slodkowicz-Kowalska, 2009).

Based on discriminative nucleotide sequences in the *E. bieneusi* ITS region, more than 90 genotypes have been identified in vertebrate hosts, and the majority of the genotypes exhibit non-host specificity. Genotyping of *E. bieneusi* using the ITS region is currently considered the most suitable molecular tool for analysis of genotype distribution and host specificity. So far, Genotype D has been reported as the most predominant genotype in most studies (Thellier and Breton, 2008).

In our study, transplantation patients only harbored genotype D. Other studies have demonstrated genotype D in transplant recipients (Galvan et al., 2011; Liguory et al., 1998). In one study, however, Genotype C was reported as the predominant genotype in transplant recipients (Sing et al., 2001). In Iran, only genotype D has been identified in transplantation patients so far (Agholi et al., 2013b). The reason for this observation is not clear to us, but our finding besides other studies indicates that the report of genotype D in most of patients with different types of immunity failures could be related to the widespread nature of this genotype. On the other hand, the number of transplantation patients registered in our study was lower than that of the two other groups, and more samples being available for the transplantation patients likely increased the chance of other genotypes being present in these groups of patients.

HIV-positive patients harbored both genotypes D and E. Several genotypes have been reported from HIV-positive patients (Leelayoova et

Fig. 2. Phylogenetic analysis of ITS nucleotide sequences of *E. bieneusi* isolates recovered from immunocompromised patients in Iran. The tree was constructed using the Maximum-Likelihood test and the Tamura 3-parameter model as implemented in the MEGA6 software. The numbers above the branches indicate the percentage of bootstrap samplings. Branches without numbers have bootstrap values less than 75%. Apart from the sequences generated in the present study, which are marked by black triangles, a number of reference sequences from other studies were included in the phylogenetic analysis. *Encephalitozoon cuniculi* genotype III was used as an outgroup. Abbreviations: Alg, Algeria; Arg, Argentina; Ger, Germany; PRC, People's Republic of China; USA, United States of America; Jap, Japan; Bra, Brazil.

al., 2006; Ojuromi et al., 2012; Saksirisampant et al., 2009; Stark et al., 2009; Sulaiman et al., 2003a), and it appears likely that there is no particular susceptibility to specific genotypes in this group of patients.

There are few studies on microsporidiosis in cancer patients, and most of these studies have been carried out on incidence and specific detection of microsporidia (Lono et al., 2008). To our knowledge, this study is one of the first characterizing *E. bieneusi* genotypes in cancer patients, and we identified both genotypes D and E. Undoubtedly, cancer patients make up a group susceptible to microsporidiosis, in which *E. bieneusi* is the most prevalent species.

To date, only two studies have been performed in Iran on HIV-positive patients and liver transplant patients, revealing the presence of genotypes D and K in these patients (Agholi et al., 2013a, b). Genotype E has been reported in HIV-positive patients only in South America (Peru [Sulaiman et al., 2003a]), and Southeast Asia (Vietnam [Espern et al., 2007] and Thailand [Leelayoova et al., 2006]), and there has been no reporting of this genotype in humans in the Middle East; however, with our data at hand, it appears that Genotype E has a broader distribution. More genotyping studies on *E. bieneusi* are needed to more clearly map the distribution, main sources, and transmission route of this genotype.

Genotype E appears to be characterized by non-host specificity, having been reported from animals and, more recently, bird (Accession no. AB897499) as well as humans (Espern et al., 2007; Leelayoova et al., 2006; Sulaiman et al., 2003a).

Genotype E is synonymous with “Ebc”, “WL13”, and “Peru 4” (Thellier and Breton, 2008) and was reported from swine in Switzerland (Breitenmoser et al., 1999), and other animals such as muskrat, racoon, beaver, fox, and otter from the US (Sulaiman et al., 2003b). According to GenBank, Genotype E was also identified from exotic birds in Iran. Therefore, birds and other animals might represent a potential source of genotype E infection.

Based on phylogenetic analysis, genotypes D and E clustered into separate groups, nesting among isolates from human, mammals, and birds. Most of the genotype D isolates clustered with strains that were isolated from cattle, horses, and birds. This finding indicates that zoonotic transmission is likely one of the most plausible routes of transmission of intestinal microsporidiosis due to this genotype. Some of our genotype D isolates (3 transplant recipients and 1 HIV+/AIDS patient) segregated to more distant positions compared with other genotype D isolates (Fig. 2), reflecting the diversity across the ITS region in our isolates.

All genotype E samples obtained in this study segregated into groups separate from pig and bird isolates, suggesting that human—rather than animals—sources are probably more important in the transmission of this genotype to humans in Iran.

5. Conclusion

Genotype D is the most frequent genotype of *E. bieneusi* identified in immunocompromised patients in Iran, and the geographical distribution of genotype E is likely to be wider than previously expected. We were not able to establish any clear relationship between specific genotypes and types of immunodeficiency, despite the fact that some genotypes were more frequently isolated from specific patients.

Conflict of interest

The authors declare that they have no conflicts of interest.

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